you need it to have an accurate estimate of the current probability to get a false positive.

4.6.8 Estimating Accuracy

In fact, our last task is to provide a method to estimate the probability of a false positive based on current status of the filter, i.e. on the number of elements currently stored in the filter, compared to its maximum capacity.

As we will see in section 4.10, this probability is roughly[12]:

$$p = \left(1 - e^{-\frac{numHashes \cdot size}{numBits}}\right)^{numHashes}$$

Listing 4.12 briefly illustrates the pseudocode that computes this method.

Listing 4.12 Method falsePositiveProbability

```
function falsePositiveProbability()
    return pow((1 - pow(E, this.numHashes * this.size /
this.numBits)), this.numHashes)
```

This concludes current section about how to implement a Bloom filter; before we delve into the theoretical part, explaining the mathematical basis of this data structure, let's first review some of the many applications for Bloom filters.